$$
\begin{aligned}
y &= A_\theta \eta + o_\theta + \epsilon, \\
\widetilde{y} &= C_\epsilon^{-1/2}(y - o_\theta), \quad B_\theta = C_\epsilon^{-1/2} A_\theta, \\
Q_\theta &= \|(I - B_\theta B_\theta^+)\widetilde{y}\|^2.
\end{aligned}
$$